CSE 422 Lab Project : Customer Churn Prediction

A Comparative Study of Classification and Clustering Models

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***Abstract*—Finding out which customers are likely to churn is crucial for a business. In this project, a dataset of 1,500 customers was analyzed to predict whether a customer will churn or not. After careful exploratory data analysis and preprocessing, three supervised machine learning models—Logistic Regression, K-Nearest Neighbors (KNN), and Neural Network—and one unsupervised machine learning model, K-Means, were applied to develop models for future predictions. After training, the performance of the models was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC curves. The results show that all three supervised models perform close to the random-guessing level, with Logistic Regression achieving the best accuracy, indicating that the available features carry very little predictive signal for customer churn.**

***Index Terms*—Customer Churn, Machine Learning, Logistic Regression, KNN, Neural Network, K-Means Clustering, Churn Prediction**

I. Introduction

CUSTOMER churn classification refers to finding out the customers who have discontinued their relationship with the company or service. Predicting potential churned customers before they discontinue their relationship gives the opportunity for the company to take proactive measures to hold their valuable customers. This allows offering discounts, personalized offers, or improved service, which is typically far cheaper than finding new customers.
The aim of this project is to build and compare machine learning models that can predict from customer past behavior, whether a given customer will churn (Churn = 1) or not (Churn = 0).

The motivation behind this project is to find hidden connections among the features using the complete machine learning pipeline—from a raw, noisy tabular dataset—and to compare how a linear model (Logistic Regression), an instance-based model (KNN), and a Neural Network behave on the same churn-prediction task. An unsupervised K-Means clustering analysis is also performed to find out how this dataset would perform if it was not labeled.

II. Dataset Description

The dataset used in this project contains 1500 data points and 9 features, including the target feature. The target feature is Churn. It is a binary variable which has marked churned customers with 1 and not churned customers with 0.

The features are summarized in Table I.

TABLE I
Description of Dataset Features

Feature	Type	Unique Values
Age	Numerical (float64)	52
Income	Numerical (float64)	1425
Gender	Categorical (str)	3 (Female, Male, Other)
Marital_Status	Categorical (str)	3 (Married, Single, Divorced)
Credit_Score	Numerical (float64)	1425
Num_Purchases	Numerical (int64)	15
Membership_Years	Numerical (int64)	14
Device_Used	Categorical (str)	3 (Desktop, Mobile, Tablet)
Churn (target)	Binary Categorical (int64)	2 (0, 1)

Out of the 9 features, 3 are categorical (Gender, Marital_Status, Device_Used) and 6 are numerical, including the binary target Churn.

III. Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of the dataset.

A. Numerical Feature Distributions

Density and histogram plots (Fig. 1) show that Age, Income, Credit_Score, Num_Purchases, and Membership_Years are all approximately symmetrically distributed, which later justified imputing missing values using the mean.

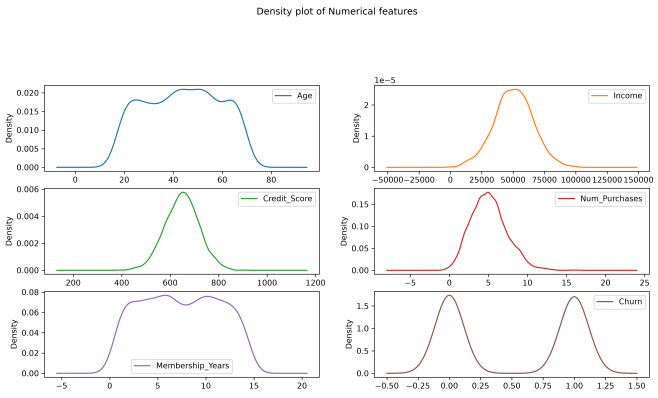


Fig. 1. Density plots of the numerical features.

Boxplots (Fig. 2) were used to inspect the numerical features.

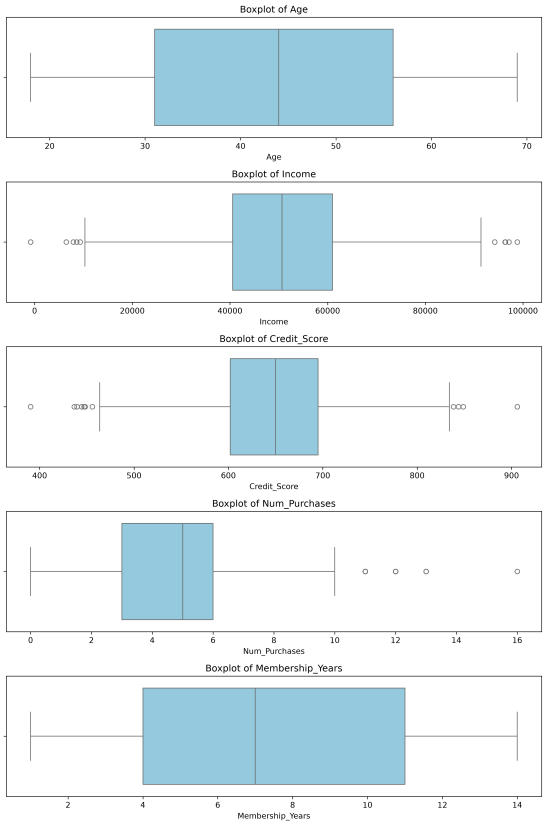


Fig. 2. Boxplots of numerical features.

From the boxplots, a single invalid negative value was discovered in the Income feature, which is not practically meaningful for an income feature.

B. Correlation Analysis

A correlation heatmap (Fig. 3) was generated for all numerical features against the target.



Fig. 3. Correlation heatmap of numerical features.

All correlation coefficients with Churn are close to zero, indicating that no numerical feature has a meaningful linear relationship with churn. No pair of features are highly correlated with each other either, so multicollinearity is not a concern for this dataset.

C. Categorical Feature Distributions

The categorical features Gender, Marital_Status, and Device_Used were all found to be well balanced across their categories (Fig. 4).

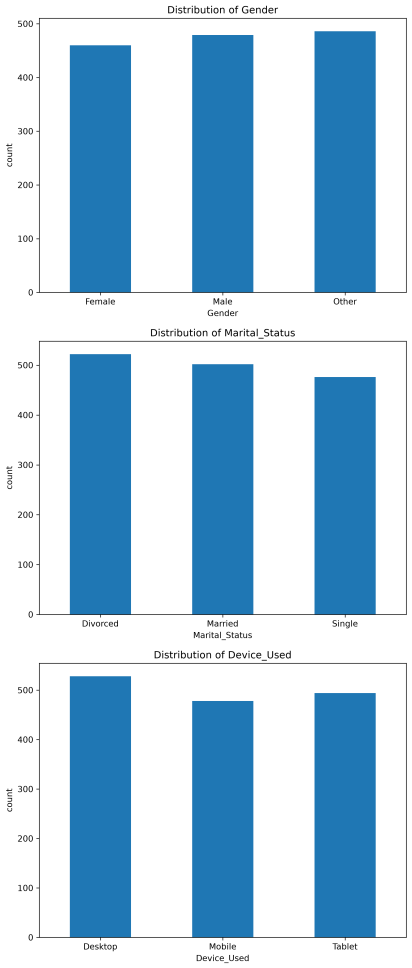


Fig. 4. Distribution of Gender, Marital Status, and Device Used.

Further analysis on Device_Used showed churn rates of Desktop, Mobile, and Tablet are very close which later motivated to group of Mobile and Tablet into a single Portable category.

IV. Dataset Pre-processing

A. Problem: Invalid and Missing Values

As identified during EDA, the dataset had two main data quality issues: one row contained a negative, practically invalid income value, and several features contained missing (null) values.

The single row with negative income was dropped, reducing the dataset from 1500 to 1499 rows. A duplicate check

confirmed there were no duplicate rows. Missing-values were then handled, *after* splitting the data into training (80%) and testing (20%) sets to avoid data leakage:

- Age, Income, and Credit_Score: Since these numerical features were all found to be approximately symmetric (skewness close to 0), missing values were imputed with the mean of the training set using SimpleImputer.
- Gender: Missing values were imputed with the most frequent category (mode) of the training set.

In both cases, the imputer was fit only on the training set and then applied to transform both the training and test sets, preventing data leakage from the test set.

B. Encoding Categorical Values

Before starting encoding, the two rarer classes of Device_Used (Mobile and Tablet) were merged into a single category - Portable, reducing it to a binary feature (Desktop vs. Portable). All categorical features (Gender, Marital_Status, Device_Used) were then encoded using One-Hot Encoding (with drop='first' to avoid the dummy variable trap), fit on the training set and applied to both sets.

C. Feature Selection

No features were dropped for feature selection, since the correlation analysis showed no redundant or highly correlated numerical features, and all categorical features were judged relevant to the problem.

D. Feature Scaling

All features (numerical and one-hot encoded) were standardized using StandardScaler, fit on the training set only and applied to both the training and test sets. Scaling is particularly important for distance-based models such as KNN and for gradient-based optimization in the Neural Network.

After pre-processing, the training set contained 1199 samples and the test set contained 300 samples, each with 10 features.

V. Model Training

Three supervised classification models and one unsupervised clustering model were trained on the pre-processed data.

A. Logistic Regression

A Logistic Regression classifier with max_iter=10000 was trained on the scaled training data as a baseline linear model.

B. K-Nearest Neighbors (KNN)

A grid search over $k \in \{1, \ldots, 20\}$ using 3-fold cross-validation was performed to select the optimal number of neighbors. The optimal value was found to be $k = 1$, and the final KNN model was trained with this setting.

C. Neural Network

A Neural Network was built using tensorflow.keras Sequential with the following architecture:

- Dense layer, 16 units, ReLU activation
- Dense layer, 8 units, ReLU activation
- Dense (output) layer, 1 unit, Sigmoid activation

The model was compiled with the adam optimizer and binary_crossentropy loss, and trained for 100 epochs with a batch size of 32.

D. K-Means Clustering (Unsupervised)

To explore the dataset without using the Churn label, all pre-processed samples (training + test, 1499 rows) were combined and scaled, and K-Means clustering was applied. The Elbow method (Fig. 5) did not show a clear elbow point, so the Silhouette score was computed for $k = 2$ to 29 (Fig. 6) to select the best number of clusters. The highest silhouette score (≈ 0.145) was obtained at $k = 20$; however, $k = 11$ was selected as a practical trade-off. This also indicates that no strong clustering was possible in the data.

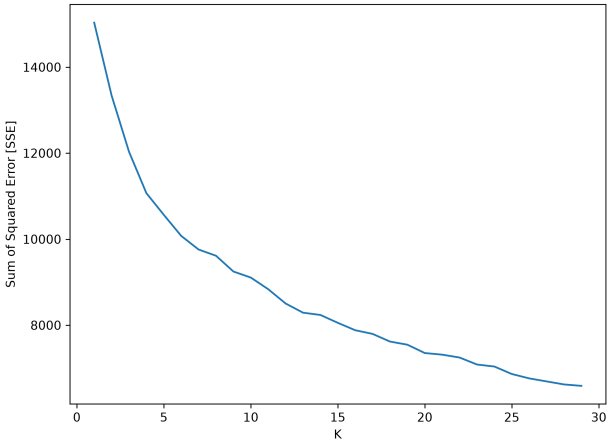


Fig. 5. Elbow method: SSE vs. number of clusters k.

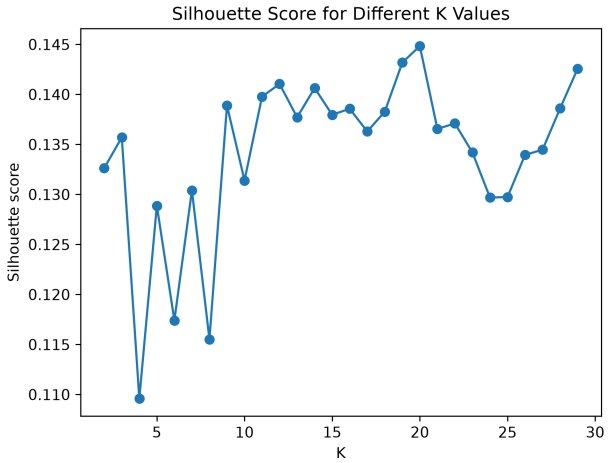


Fig. 6. Silhouette score vs. number of clusters k.

Fig. 7 visualizes the resulting 11 clusters projected on the Income and Credit_Score axes. The clusters overlapped, confirming that the customer base does not separate into strongly distinct segments.

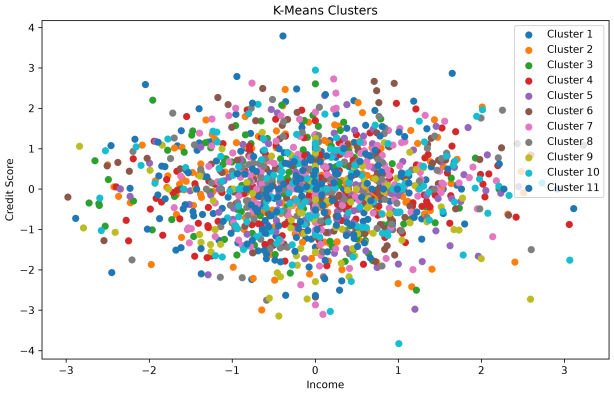


Fig. 7. K-Means clusters ($k = 11$) projected on Income vs. Credit Score.

VI. Model Evaluation

A. Confusion Matrices

Fig. 8 shows the confusion matrices for the three supervised models on the 300-sample test set.

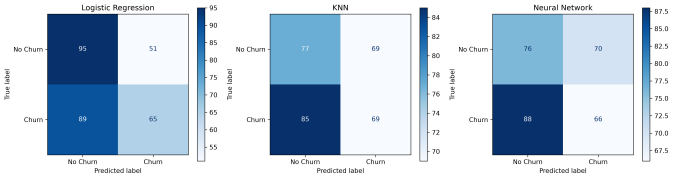


Fig. 8. Confusion matrices for Logistic Regression, KNN, and Neural Network.

B. Accuracy, Precision, Recall, F1-Score

Table II summarizes the weighted-average classification metrics for each model.

TABLE II
Classification performance on the test set

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.53	0.54	0.53	0.53
KNN ($k = 1$)	0.49	0.49	0.49	0.49
Neural Network	0.47	0.47	0.47	0.47

C. ROC Curve and AUC Score

Fig. 9 shows the ROC curves of all three models along with the random-guess diagonal.

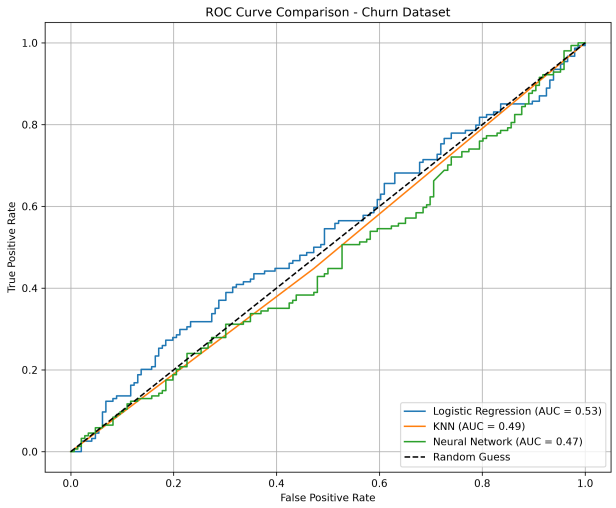


Fig. 9. ROC curves comparing Logistic Regression, KNN, and Neural Network.

All three ROC curves lie very close to the diagonal random-guess line, consistent with the low accuracy values, confirming that none of the models can meaningfully separate churners from non-churners using the available features.

VII. Conclusion

Based on the evaluation metrics, Logistic Regression achieved the best (but still weak) performance among the three supervised models, with 53% accuracy and the highest F1-score. KNN and the Neural Network performed at or below the level of random guessing (49% and 47% accuracy respectively).

The most probable reason behind these results is that, as shown clearly by the correlation analysis, none of the available features (Age, Income, Credit_Score, Num_Purchases, Membership_Years, Gender, Marital_Status, Device_Used) has any meaningful statistical relationship with the target feature Churn. Correlations were all near zero, and group-wise means of numerical features were nearly identical between churned and non-churned customers. This suggests that the dataset either lacks the features that actually drive churn or that churn in this particular dataset is largely driven by random/unobserved factors. The unsupervised K-Means analysis reinforces this conclusion: no clear elbow point and consistently low, similar silhouette scores across a wide range of k indicate that the customer base does not separate into well-defined natural segments either. Future work could focus on collecting additional behavioral features (e.g., support tickets, usage frequency, billing history) that may carry stronger predictive signal for churn.